ENGINEERING MATHEMATICS-II

Department of Science and Humanities

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UNIT 1: INTEGRAL CALCULUS

Class 8:applications of triple integral on finding average value USING TRIPLE INTEGRAL

AVERAGE VALUE OF A FUNCTION OF THREE VARIABLES

Recall that we found the average value of a function of two variables by evaluating the double integral over a region on the plane and then dividing by the area of volume of the solid.

If is integrable over a solid bounded region E with positive volume *V(E)*

AVERAGE VALUE OF A FUNCTION OF THREE VARIABLES: EXAMPLE 1

Note that the volume is

Suppose the temperature at a point is given by . Find the average temperature in the cube with opposite corners at (0,0,0) and (2,2,2).

In two dimensions:

1. Add up the temperature at each point in a region
2. Divide by the area.

In three dimensions:

3. Add up the temperature at each point in space.
4. Divide by the volume.

AVERAGE VALUE OF A FUNCTION OF THREE VARIABLES: EXAMPLE 1 CONTD…

Therefore, the average temperature in the cube is

AVERAGE VALUE OF A FUNCTION OF THREE VARIABLES: EXAMPLE 2

The temperature at a point of a solid E, bounded by the coordinate planes and the plane is Find the average temperature over the solid.

Use the theorem given above and the triple integral to find the numerator and the denominator. Then do the division.

Notice that the plane has intercepts (1,0,0),(0,1,0) and (0,0,1).

AVERAGE VALUE OF A FUNCTION OF THREE VARIABLES: EXAMPLE 2 contd....

The region E looks like

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Hence the triple integral of the temperature is

$$\iiint_E f(x,y,z)\,dV = \int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y}(xy+8z+20)\,dz\,dy\,dx = \frac{147}{40}.$$

The volume integral is

$$V(E) = \iiint_E 1\,dV = \int_{x=0}^{x=1}\int_{y=0}^{y=1-x}\int_{z=0}^{z=1-x-y} 1\,dz\,dy\,dx = \frac{1}{6}.$$

Hence the average value is

$$T_{ave} = \frac{147/40}{1/6} = \frac{6(147)}{40} = \frac{441}{20}\ {}^\circ\mathrm{C}$$

AVERAGE VALUE USING TRIPLE INTEGRAL: EXAMPLE:1

Find the average value of the function $f(x, y, z) = ye^{-xy}$ over the rectangular prism $0 \leq x \leq 2, 0 \leq y \leq 3, 0 \leq z \leq 2$

THANK YOU